

# **ASSESSMENT TOOL 2 -SCENARIO BASED**

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## Q1. Error Detection & Correction — “Catch the Corrupted Message”:

SOLN:

```
import random

def binary(text):

    return ''.join(format(ord(c), '08b') for c in text)

def parity(bits):

    return bits.count('1') % 2

def checksum(text):

    return sum(ord(c) for c in text)

def flip(bits):

    i = random.randint(0, len(bits)-1)

    return bits[:i] + ('1' if bits[i]=='0' else '0') + bits[i+1:]

messages = ["HELLO", "WORLD", "PYTHON", "DATA", "NETWORK"]

for msg in messages:

    b = binary(msg)

    p = parity(b)

    c = checksum(msg)

    rb = flip(b)

    rp = parity(rb)

    rmsg = ''.join(chr(int(rb[i:i+8],2)) for i in range(0,len(rb),8))

    rc = checksum(rmsg)

    print("\nMessage:", msg)

    print("Parity:", "ERROR" if p != rp else "OK")

    print("Checksum:", "ERROR" if c != rc else "OK")
```

#### OUTPUT:

Message: HELLO

Parity: ERROR

Checksum: ERROR

Message: WORLD

Parity: ERROR

Checksum: ERROR

Message: PYTHON

Parity: ERROR

Checksum: ERROR

Message: DATA

Parity: ERROR

Checksum: ERROR

Message: NETWORK

Parity: ERROR

Checksum: ERROR

=== Code Execution Successful ===

#### OBSERVATION:

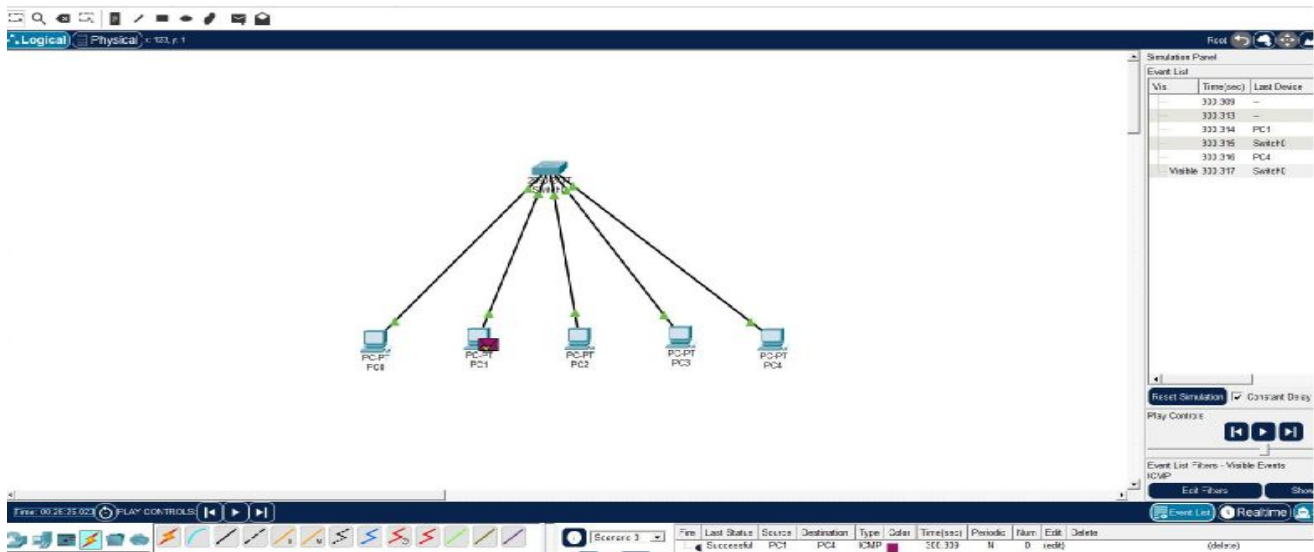
The transmitted messages were intentionally corrupted by flipping one random bit. Both the parity bit and checksum detected the errors successfully. The experiment shows that checksum is generally more reliable than parity for detecting transmission errors.

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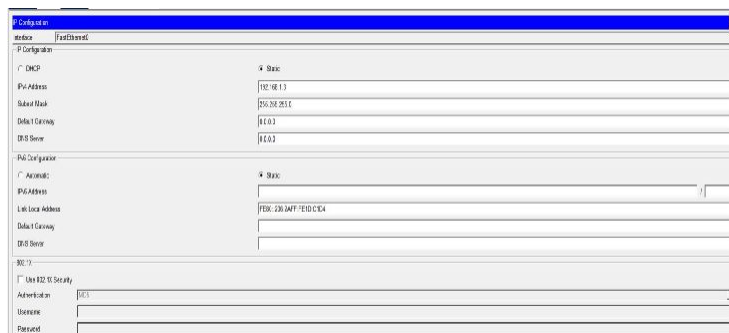
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## 2. LAN & Multiple Access — “Set Up and Test a Mini Home/Lab Network”:

This is topology:



This is IP configuration:



This is ping test:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

OBSERVATION:

During the experiment, all five PCs connected through the **switch** communicated successfully using ping. The switch forwarded data only to the destination PC, reducing unnecessary traffic. When the switch was replaced with a **hub**, data was broadcast to all connected PCs, increasing network traffic and the chance of collisions. This showed that switches provide better performance and efficiency than hubs. Therefore, switches are preferred in modern LANs.

Q3: WIRELESS BASIC “Survey Your Own Wi-Fi World”

| Sample Wi-Fi Comparison Table |                 |         |                    |
|-------------------------------|-----------------|---------|--------------------|
| SSID                          | Signal Strength | Channel | Wi-Fi Standard     |
| Home_WiFi                     | -40 dBm         | 36      | Wi-Fi 6 (802.11ax) |
| Airtel_Xstream                | -55 dBm         | 6       | Wi-Fi 5 (802.11ac) |
| JioFiber                      | -60 dBm         | 11      | Wi-Fi 5 (802.11ac) |
| Hostel_WiFi                   | -70 dBm         | 1       | Wi-Fi 4 (802.11n)  |
| TP-Link                       | -75 dBm         | 44      | Wi-Fi 6 (802.11ax) |

Observation

The Wi-Fi scan showed that nearby networks have different signal strengths and Wi-Fi standards. Networks with stronger signals (closer to **-40 dBm**) provide better performance than weaker signals (around **-70 dBm**). Wi-Fi 6 networks offer higher speed, lower delay, and better performance in crowded areas compared to older Wi-Fi standards.

Conclusion

Wi-Fi 6 is more efficient than older Wi-Fi versions because it supports more devices at the same time with better speed and stability. Upgrading to Wi-Fi 6 is beneficial in places like classrooms, hostels, and offices where many users connect simultaneously.